



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KARPAGA VINAYAKA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

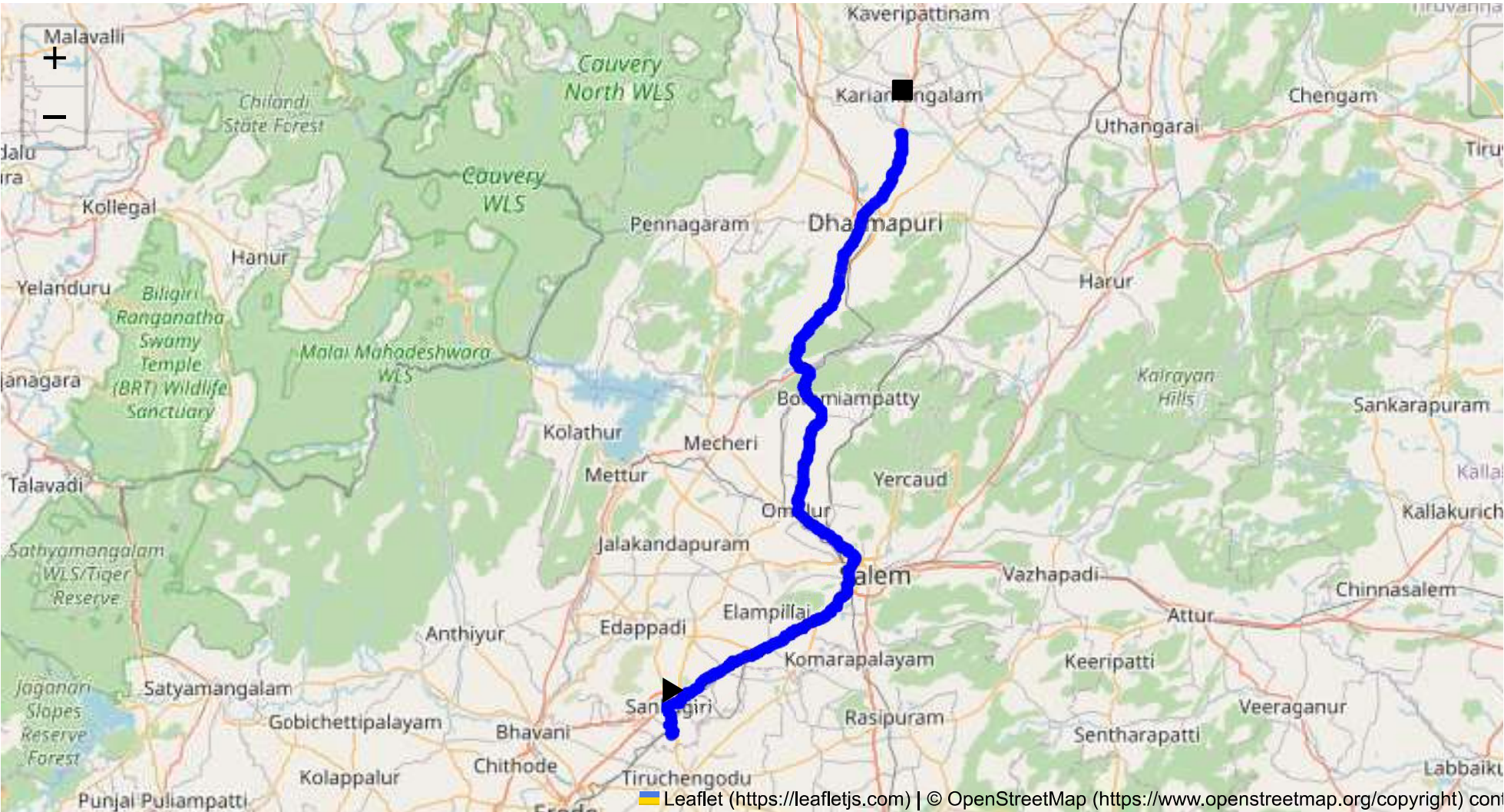
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 120.47 km
Estimated Duration: 2.2 hours
Adjusted Duration (Heavy Vehicle): 2.8 hours
Start: (11.4381, 77.8734)
End: (12.254624, 78.193633)



Welcome to the Journey Risk Management Study

- 1. **Overview of the Route Map:** The route from CVQF+23W, Sangagiri to 753V+QFC, Periyanaahalli spans approximately 120.47 kilometers, predominantly following regional highways that connect these areas. The route transitions through several smaller towns and rural sections, providing a mix of open highway segments and areas with denser local traffic.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:** This region of Tamil Nadu experiences a tropical climate with high temperatures year-round. The monsoon season, lasting from June to September, can bring heavy rains, potentially leading to flooding and reduced visibility. Heat waves during the summer months can affect vehicle performance.
3. **Analysis of Traffic Patterns:** Traffic is generally moderate, with peak hours typically between 8:00-10:00 AM and 5:00-7:00 PM. Congestion can occur near town centers and junctions, particularly in Sangagiri and any major intersections with national highways.
4. **Assessment of Road Quality and Infrastructure:** The route predominantly consists of well-paved roads, though some rural segments may have less maintained surfaces. Potholes and uneven patches are common in these areas, which can be particularly hazardous in wet conditions.
5. **Suggestions for Alternative Routes for Emergencies:** If the main route is obstructed, consider taking National Highway 44 (NH44) which offers a more direct path with better infrastructure. However, this might increase travel time depending on specific detours taken.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:** Regulations require trucks carrying hazardous materials to display appropriate signage and travel during designated hours on certain stretches. Strict adherence to loading limits and possession of necessary documentation are mandatory.
7. **Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:** Historical data indicates occasional incidents involving heavy vehicles, predominantly due to adverse weather conditions and driver fatigue. There have been few reported incidents involving hazardous materials, mostly related to poor handling or improper storage.
8. **Environmental Considerations and Sensitive Areas:** The route passes through several agricultural areas, necessitating caution to avoid spills or contamination. Additionally, nearby water sources and wildlife areas require careful monitoring to prevent environmental damage.
9. **Analysis of Communication Coverage:** Communication coverage is generally reliable along major sections of the route, though some rural areas may present sporadic dead zones due to limited mobile network reach.
10. **Estimated Emergency Response Times for Different Route Segments:** Response times vary, with urban areas like Sangagiri generally better equipped, resulting in quicker response times (approximately 15-30 minutes). Rural sections may see delayed response times, extending up to 45-60 minutes depending on remoteness and accessibility.
11. **Overall Summary of Risk Assessment:** The route from Sangagiri to Periyanaahalli, while generally safe under normal conditions, presents hazards related to weather, road conditions, and potential congestion points. Truck drivers, especially those transporting hazardous materials, should remain vigilant during the monsoon season and plan trips outside peak hours to reduce risks. Regular vehicle maintenance and adherence to safety protocols are advised to mitigate potential hazards. Alternative routes, emergency contacts, and environmental considerations should be factored into pre-trip planning to ensure a safe journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	Medium	11.45000, 77.87213	30 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.45309, 77.87325	30 KM/Hr
6	Turn	Medium	11.46400, 77.87323	30 KM/Hr
7	Turn	Medium	11.46765, 77.87037	30 KM/Hr
8	Turn	High	11.47403, 77.86841	15 KM/Hr
9	Turn	High	11.49149, 77.89601	15 KM/Hr
10	Turn	High	11.49224, 77.89601	15 KM/Hr
11	Turn	Medium	11.53512, 77.95717	30 KM/Hr
12	Turn	Medium	11.53606, 77.95869	30 KM/Hr
13	Turn	Medium	11.64183, 78.11810	30 KM/Hr
14	Turn	Medium	11.64338, 78.11998	30 KM/Hr
15	Turn	Medium	11.91357, 78.05868	30 KM/Hr
16	Turn	Medium	11.92556, 78.06499	30 KM/Hr
17	Turn	Medium	11.93694, 78.05425	30 KM/Hr
18	Turn	Medium	11.95531, 78.04698	30 KM/Hr
19	Turn	Medium	11.95854, 78.04990	30 KM/Hr
20	Turn	Medium	11.96015, 78.04945	30 KM/Hr
21	Turn	Medium	11.96370, 78.05142	30 KM/Hr
22	Turn	Medium	11.96433, 78.05113	30 KM/Hr
23	Turn	High	11.96542, 78.04938	15 KM/Hr
24	Turn	High	11.96650, 78.04927	15 KM/Hr
25	Turn	Medium	11.96804, 78.05080	30 KM/Hr
26	Blind Spot	Blind Spot	12.25427, 78.19648	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
23	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Global Medical Centre Jagirammapalayam	11.6894409, 78.1108847	30 km/h	Medium
30	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
33	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
34	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
36	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
37	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
39	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
41	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
42	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
43	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium